

Selective Entry High School (SEHS) Maths Games Day – 2026

Problem Solving Booklet

Ernest LIN

Yin Chun WONG

Maths Extension Group (MEG)

Melbourne High School

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Questions

a)

The line $y + 2x - 4 = 0$ passes through point M with coordinates $(a, 6)$ and point N with coordinates $(3, b)$

i) Find the values of a & b (1)

Another line passes through point M but is perpendicular to the 1st line. It also passes through point O with coordinates (c, d) such that MO has a distance of $5\sqrt{5}$

ii) Find the 2 different sets of solutions to the coordinates of point O. (3)

iii) Given that point O lies on the left side of the y-axis, find the area of the triangle MNO (2)

iv) Considering that the points M, N, O all lie on a circle, find the coordinates of the centre of the circle (1)

b)

Lines α & β both pass through point R. Line α is given by the equation $y - 3x = 3$, passes through point P with coordinates $(10, a)$ and ends point Q which is also its y-intercept. Point R splits PQ in the ratio of $PR:RQ = 3:2$

i) If line β is perpendicular to line α , find the equation for line β in general form (3)

ii) Point S lies above the x-axis so that QS is parallel to line β & $12\sqrt{10}$ in magnitude. Find the coordinates of S (4)

iii) If Points M & N lie on lines α & β respectively so that the line MSN has the equation $x - r = 0, r \in \mathbb{R}$, find the area of triangle MRN in the form of $\frac{k}{3}$, where $k \in \mathbb{N}$ (3)

c)

Point A has the coordinates $(-4, -10)$. Point B is located from point A with an angle of elevation of 71.57° , and the distance between the 2 points is $7\sqrt{10}$ units

i) Find the coordinates of point B (to the nearest integers) (2)

ii) Find the equation of the line passing through point B and perpendicular to the line linking A & B. Give the equation in general form (2)

Points A & B are both on the circumference of a circle with a radius of $\sqrt{185}$ units. Point O is the centre of this circle

iii) If point O lies on the right of the y-axis, find the coordinates of point O (4)

iv) Find the area of the sector AOB, giving your answer to 3 d.p. (5)

Relations

(/35)

a)

The variable a relates to time (t) such that the rate of change of a (relative to t) is always equal to the value of a multiplied by e^t at any given time

i) Given that $a = e$ when $t = 0$, find the equation for the relationship & solve for a (3)

ii) Find a'' in terms of t only, expressing your answer in the form $e^{f(t)} + e^{g(t)}$ (3)

iii) A new variable b is given by the equation $b = \frac{2f(t)}{g(t)}$ (functions from part iii). Find the time that b is at a value of 2 (3)

b)

Variable y is related to x with the relation $e^{\frac{y}{x}} = x^2 - 2$. Given that x changes by 4%, find the percentage change in y when $x = 2$ in exact value. (5)

Hint: $\Delta y\% = \frac{\Delta y}{y} \times 100$

c)

Variable p relates to q such that $p = 2^{(q+1)}$, while variables r & s are related by the equation $r = 5 \log(s + 3)$. When r is plotted against $\sin^2 p$ (p is in radians), a straight-line graph is created, passing through points $(0, 5)$ & $(1, 10)$. Find the value of s when q is 2 (to 2 d.p.) (5)

d)

The sector of a circle has angle x and radius r , such that the radius relates to the angle by the rule $r = x^2 - \frac{1}{x}$. If the radius of two of these special sectors differs by 10%, and one sector has an angle of 1.2 (radians), find the percentage difference in the chord length to 4 s.f. (6)

Hint: $\text{Chord length} = 2r \sin\left(\frac{x}{2}\right)$

e)

The inverse of a satellite's signal strength (S) consistently deviates from some proportional value of the square of the satellite's altitude in km (h^2) by a constant value.

i) Considering that the satellite strength is 0.2 units when the altitude is 3 km, and doubles when the satellite flies at an altitude 1 km lower, find $S = f(h)$ (4)

ii) The altitude (h) varies with the orbital angle θ (in radians) according to the relationship $\ln(h) + e^{-2h} = \sin \theta$. Find the exact value of the rate of change of signal strength with respect to the orbital angle when the satellite is flying 1000 m above ground. Express your final answer in exact form as a single fraction. (6)

Hint: Use appropriate small angle approximation

a)

Prove the identities below

Hint: Start with LHS for all 3 identities

i)

$$\sec(\theta) - \frac{\tan(\theta)}{\csc(\theta)} = \cos \theta \quad (2)$$

ii)

$$\left(\frac{1}{\cos(x)} - \tan(x) \right)^2 = \frac{1 - \sin(x)}{1 + \sin(x)} \quad (3)$$

iii)

$$\frac{\cos(3x)}{\sin(x)} + \frac{\sin(3x)}{\cos(x)} = 2\cot(2x) \quad (4)$$

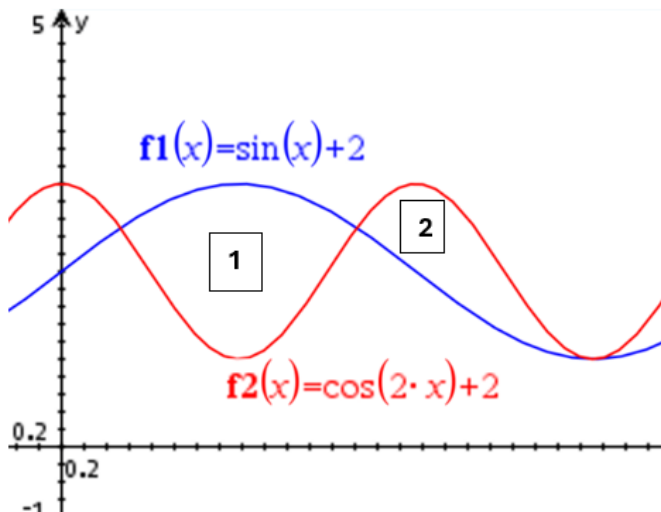
b)

Solve the equation

$$(1 + \sqrt{3}) \tan x - 1 = 4\sqrt{2} \sin x - \sqrt{3}, \quad x \in [0, 90^\circ] \quad (8)$$

Hint: Use the special angle values of sin & cos

c)



i) State the sequence of transformations (in order) required to make $f_1(x)$ into $f_2(x)$ (1)

ii) Find the coordinates of all three points in the diagram where the graphs intersect, giving the coordinates in exact value (6)

iii) Find the difference between the volumes of the solids generated by rotating each graph 225° about the x-axis, bound by the lines $x = 0$ & $x = \frac{\pi}{4}$, given to 2 d.p. (9)

iv) Find the ratio between the areas labelled 1 & 2 (areas bound by 2 graphs), giving your answer in the form $a:b$ (5)

d)

Find the **number** of solutions to the equation

$$\frac{\sin(x) + \sin(2x) + \sin(3x) + \sin(4x)}{\cos(x)} = 0$$

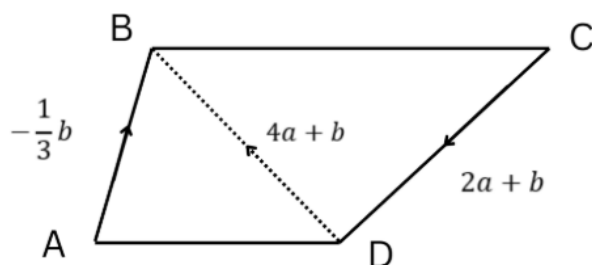
Where $0^\circ \leq x < 360^\circ$ (7)

HINT: Use the same identity/logic repeatedly

Vectors

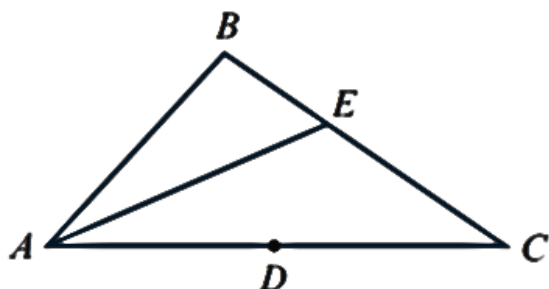
(/32)

a)



- i) Using the given vectors, prove that the quadrilateral ABCD is a trapezium (2)
- ii) Find the vector connecting point C and the intersection of the lines AC & BD in terms of a & b in exact value & expanded form (5)

b)



- i) Given that $\vec{AD} = \frac{1}{2}\vec{AC} = x$, $\vec{AB} = y$, and that $\frac{BE}{EC} = \frac{1}{z}$, find AE in terms of vectors x, y & number z , expressing your answer in a single fraction (3)
- i) Point F forms lies on the intersection of extended line AB & extended line DE. If $\vec{DE} : \vec{DF} = 1 : 8$, find $\vec{AF}$ (7)

Hint: Answer should be given in terms of y only

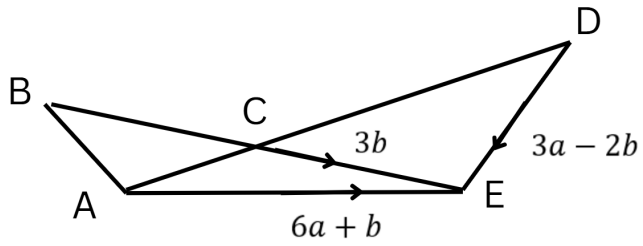
c)

Vectors $\vec{AX}$ & $\vec{AY}$ both face the positive x direction, forming a 70° angle at point A. Point X lies on the ground, and Point Y is above ground such that the line XY line is perpendicular to the ground. Vectors $\vec{BX}$ & $\vec{BY}$ both face the negative x direction, are perpendicular with each other, and $3|\vec{BY}| = 2|\vec{BX}|$

Given that $\vec{AX} = 9i - 2j$, find $\vec{AB}$ in the form $ai + bj$, where a & $b \in \mathbb{R}$ (to 4 d.p.) (7)

Hint: Do not use dot product or vector angle formulas, & ensure to draw a diagram to assist with your calculations

d)



To clarify, $\overrightarrow{BE} = 3b$

Point X lies on $\overrightarrow{AE}$ so that $\overrightarrow{AB}$ is parallel to $\overrightarrow{CX}$.

By first finding $\overrightarrow{AC}$, find the vector of $\overrightarrow{AX}$ in terms of vectors a & b , giving your answer in the form $k \left(a - \frac{b}{n} \right)$, where k, m & n are natural numbers (8)

Logarithms

(/47)

a)

Solve the equations

i) $\log_4 x - \log_x 32 = -3 \log_{16} 64$ (5)

ii) $\log_x \frac{1}{2} - \log_{\frac{1}{\sqrt{x}}} \frac{2}{x} = -1$ (4)

b)

Solve the following pairs of simultaneous equations

i) (6)

$$\log_4 a + 3 \log_8 b = \frac{5}{2}$$

$$2^a = \frac{16^4}{4^{b^2}}$$

ii) (3)

$$\log_2(xy^2) = 0$$

$$\log_2(x^2y) = 3$$

c)

A man is driving down a straight road in his car when he suddenly spots an obstacle. He attempts to break the car.

- The thinking distance (initial distance not braking due to reaction time) is modelled by the formula $9 \ln(t^4)$, where $t > 0$
- The braking distance is modelled by the formula $9 \log(t)$, where $t > 0$

Using the information, now find:

- i) The total stopping distance, expressed in the form of $a \ln(t)$, where a is rounded to an integer value (4)
- ii) The exact time when the velocity first goes below two times the time value (you may use the approximation given in part i) (3)
- iii) The range of time for which the magnitude of the deceleration of the car is greater than the velocity it experiences (3)

d)

Michael, a semi-generous & lucky person, wins a sum of \$10,000 in a lottery. He decides to donate half of his money and invest the rest. He then continues to follow the “half-of-the-gain” investment-donation rule, where every time his investment **doubles**, he donates half of the **profit** he makes, while adding the other half to his current remaining investment.

By denoting each doubling of investment (a “doubling”) as k , where $k \in \mathbb{R}^+$, find:

i) The function for profit in terms of doublings (1)

ii) The function for donation in terms of doublings (1)

iii) The function for total (sum of) donation in terms of doublings, given in the form $S_k = a \left(\frac{b}{c}\right)^k$, where a , b & c are integers (2)

Hint: Make sure to account for the first/initial donation

Thus, find the time (to the nearest day) it takes for Michael to donate \$25,000 when:

iv) The market growth rate is 0.5% per day (5)

Hint: Use the compound interest formula $P = P_0 \left(1 + \frac{r}{100}\right)^t$

v) The market growth rate is 0.3%/day, and Michael wins \$15000 initially instead (4)

vi) Michael wins \$10000, and the market growth rate starts at 0.2%/day, but then switches to a growth rate of 0.3%/day after the first 100 days (6)

Hint: One of the doublings would be split between 2 growth rates. Thus, you must consider the proportion of the time of doubling spent in the first rate, known as the growth factor (G), which is given by the expression: $G = \frac{2}{\left(1 + \frac{r}{100}\right)^t}$, and hence find the remaining growth required to occur at the second rate.

The displacement of a particle is modelled by the following equation

$$8y''' + 77y'' - 512y' - 4928y = 5e^{-9t}$$

Where $y' = \frac{dy}{dt}$, $y'' = \frac{d^2y}{dt^2}$, etc.

The Laplace transform $\{\mathcal{L}(x)\}$ is given by the following equation

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

a)

i) Prove that the Laplace transform of y' is equal to the expression: $s\mathcal{L}\{y\} - y(0)$ (4)

ii) Prove that $\mathcal{L}\{y''\} = s^2\mathcal{L}\{y\} - sy(0) - y'(0)$, and using the same logic, prove that $\mathcal{L}\{y'''\} = s^3\mathcal{L}\{y\} - s^2y(0) - sy'(0) - y''(0)$ (3)

b)

The initial conditions of the equation are given.

$$y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1$$

Perform the Laplace transform on the left side of the equation & simplify. (2)

c)

i) Prove that the Laplace transform of $5e^{-9t}$ is $\frac{5}{s+9}$ (4)

ii) Hence or otherwise, find a generalised solution to the Laplace transform of all functions with the form e^{at} (1)

iii) The hyperbolic sine function is given by $\sinh(t) = \frac{e^t - e^{-t}}{2}$. Using c ii), find the generalised solution of $\mathcal{L}(\sinh kt)$ in expanded form (4)

d)

i) Summarising the above steps, express the Laplace transform of the given equation in the form $\mathcal{L}\{y\} = \frac{1}{f(s)g(s)h(s)}$ (4)

ii) Using partial fractions & knowledge from part c ii), perform the inverse Laplace transform to solve for y (3)

e)

By utilising the knowledge you learnt from part a-d (do not use other methods)

i) Find the Laplace transform of the function

$$\cos(2t) + 7 \sin(2t)$$

Expressing your answer as a single fraction (6)

ii) Solve the differential equation

$$\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + 2y = 2e^{-3x}$$

With initial conditions of $y(0) = 5$ and $y'(0) = 7$ (6)

Problems in 3D

(/32)

a)

3 forces in 3D space acting on the origin are in equilibrium

For $\vec{F}_1$ & $\vec{F}_2$:

- When projected onto the x-y plane, they form a straight line that passes through the origin, passes through quadrants 1 & 3, & makes a 30° angle with the y-axis
- $\vec{F}_1$ points in the positive y direction
- When the projected line is split into 2 parts by the origin, the two line segments are equal in length
- The 2 forces share a vertical plane (an infinite flat surface which is perpendicular to the horizontal plane, $z = 0$). If the plane is viewed head-on, the forces are perpendicular to each other
- $|\vec{F}_1| = 2 |\vec{F}_2|$

$\vec{F}_3$ points directly downwards with a magnitude of 10 N

By drawing 2 labelled diagrams of $\vec{F}_1$ & $\vec{F}_2$, find $\vec{F}_1 - \vec{F}_2$ in the form of $pi + qj + rk$, giving your answer in exact form (10)

Hint: Consider similar triangles

b)

A magnetic ball weighing 1kg is in a 3D space such that it is:

- Starting initially at point $P = (0, 0, 20)$
- Projected with an initial velocity of 2 ms^{-1} , with the direction being upwards so that it makes a 60° angle with the z-axis, and going towards the negative y direction, making a 30° angle with the x-axis

The particle is subjected to:

- Gravity (assume 9.8 for gravitational constant)
- A constant attractive magnetic force of 15N coming from the negative x direction

The particle crosses the $x = 0$ plane at point Q, at which it stops indefinitely. A line is then drawn between points P & Q to form the line PQ.

- i) Find the initial velocity vector of the particle (when $t = 0$), giving your answer in the form of $(xi + yj + zk)/t$ (8)

HINT: Use a set of 3-way simultaneous equations

ii) Find the functions modelling the particle's displacement for each of the 3 dimensions in terms of time, giving all values in exact form (5)

iii) Find the coordinates of point Q as exact value (3)

iv) Find the volume bound by the 4 surfaces created by

- The particle's path
- The line PQ
- The projections of the particle's path & line PQ on the $z = 0$ plane, creating a 2D "curtain" that encloses the whole volume

Given to 2 d.p. (6)

a)

- i) Prove the Pascal's Identity (3)

$$\binom{a+1}{b} = \binom{a}{b-1} + \binom{a}{b}$$

- ii) Prove that, by the Binomial Theorem or otherwise, the identity below (2)

$$\sum_{j=0}^n \binom{n}{j} = 2^n$$

- iii) Hence or otherwise, prove that, for all n, (6)

$$\sum_{k=1}^n k \binom{n}{k} = n \cdot 2^{n-1}$$

HINT: Expand the terms

b)

The Taylor Series expansion of a function can be given by

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)}{n!} (x-a)^n$$

- i) Prove that the Taylor Series of $\ln(1+x)$ can be generalized by the series

$$\sum_{n=1}^{\infty} a^{n-b} \frac{x^n}{cn}$$

Where a, b, & c are integers (4)

- ii) Hence or otherwise, for $0 \leq r < 1$, prove that (3)

$$\int_0^r \frac{\ln(1+x)}{x} dx = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} r^n$$

Now consider the expression I , which is given by:

$$I = \int_0^1 \frac{\ln(1+x)}{x} dx$$

iii) Show that: (5)

$$I = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Hint: Relate the alternating version to the full sum

iv) The Parseval Identity states that the square of the average value of $(f(x))^2$ over a certain interval can be given by an infinite series. The expression is denoted below

$$\frac{1}{b-a} \int_a^b (f(x))^2 dx = \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Using the values:

- $f(x) = x$
- Interval = $[-\pi, \pi]$
- $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$
- $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$

Find the exact value of I . (10)

a)

- i) In a chess tournament, every player plays every other player exactly once. No one draws. A “king” is a player who, for every other player X , either beat X directly or beat someone else who beat X . Prove that for every such tournament, the person with the most wins is King. (4)
- ii) Furthermore, prove that if no players won all their games, then there must be at least three kings. (4)

b)

Consider a series of events that happen at independent random times, where a process can be completely characterized by the probability per unit time (p) of an event happening. p is defined such that the probability of an event happening in an infinitesimal time dt equals $p dt$.

- i) Show that starting at any particular time (not necessarily the time of an event), the probability that the next event happens between t and $t + dt$ equals $e^{-pt}p dt$. (6)
- ii) Show that starting at any particular time (not necessarily the time of an event), the average (expected value) of waiting time to the next event is equal to $\frac{1}{p}$. (6)

Hint: Expected value is given by $E(x) = \int_0^{\infty} x \cdot f(x) dx$, where x is the random variable & $f(x)$ is the probability distribution function

- iii) Using the above probability distribution $e^{-pt}p dt$, prove that the average length surrounding a randomly chosen point in time is different from the expectation value of the waiting time. Hence, find the average length. (3)

c)

Define the spacetime interval between two events:

$$s^2 = t^2 - x^2$$

A Lorentz Transformation with velocity v is given by:

$$t' = \gamma(t - vx) \quad x' = \gamma(x - vt)$$

Where $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$, and let $c = 1$

- i) Prove that the spacetime interval is unchanged under Lorentz Transformations below (4)

$$t'^2 - x'^2 = t^2 - x^2$$

- ii) Let an inertial frame (a frame of reference moving with constant velocity) move with velocity u with respect to a fixed frame. Another frame moves with velocity v with respect to the frame moving with velocity u .

Prove that the composition of two Lorentz transformations is again a Lorentz transformation. (5)

- iii) Hence, find the combination of the velocities and its γ . (2)

Hint: An example of combining velocities would be a bus travelling at 30 km/h, and a person on the bus throwing a ball 20 km/h forward in the frame of the bus, resulting in a combined velocity of the ball being roughly 50 km/h.

- iv) Prove that the time dilation from the Lorentz Transformation is given by (4)

$$\Delta t' = \gamma \Delta t$$

Solutions

Geometry - Solutions

(/30)

A1)

$6 + 2a - 4 = 0, \quad a = -1$ $b + 2(3) - 4 = 0, \quad b = -2$	M1 Ans
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A2)

Original line: $y = -2x + 4$ Perpendicular line (passes through m): $y - 6 = \frac{1}{2}(x + 1), \quad y = \frac{x}{2} + \frac{13}{2}$ Given $MO = 5\sqrt{5}$ $(5\sqrt{5})^2 = 125 = (x + 1)^2 + (y - 6)^2$ Solving simultaneous questions: $= (9, 11) \text{ \& } (-11, 1)$	M1 M2 M3 Ans
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A3)

Left of y-axis, so O $\rightarrow$ choose $(-11, 1)$ Line MN: $= \sqrt{(3 + 1)^2 + (-2 - 6)^2} = 4\sqrt{5}$ Since NMO is right-angled, the area is given by $\frac{1}{2} \times 4\sqrt{5} \times 5\sqrt{5} = 50 \text{ (units)}$	M1 M2 Ans
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A4)

Since triangle MNO is right-angled, line NO is the diameter of the circle Centre of circle = midpoint of NO, given by $\left(\frac{3 - 11}{2}, \frac{-2 + 1}{2}\right) = \left(-4, -\frac{1}{2}\right)$	M1 Ans
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B1)

$a - 3(10) = 3, \quad a = 33$ Point P = (10, 33), Point Q = (0, 3) By vector or other methods $\text{Point R} = (4, 15)$ Gradient of line $\beta = -\frac{1}{3}$ $y - 15 = -\frac{1}{3}(x - 4)$ $3y + x - 49 = 0 \text{ (general form only*)}$	M1 M2 M3 Ans
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B2)

Point Q = (0, 3) Let the coordinates of S be (c, d)	
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Since QS and line β are parallel, they have the same gradient $-\frac{1}{3} = \frac{d-3}{c}, \quad c = 9 - 3d$ $c^2 + (d-3)^2 = (12\sqrt{10})^2 = 1440$ Solving simultaneous equations $d = -9, 15 \text{ or } c = \pm 36$ Since S is above x-axis, d must be positive = 15 Coordinates = $(-36, 15)$	M1 M2 M3 M4 Ans
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B3)

MSN has equation $x = -k$, meaning Vertical Line Point M & N will have SAME x-coordinates = -36 $3y + (-36) = 49, \quad y = \frac{85}{3}, \text{ Point M} = \left(-36, \frac{85}{3}\right)$ $y - 3(-36) = 3, \quad y = -105, \text{ Point N} = (-36, -105)$ $MN = \frac{85}{3} + 105 = \frac{400}{3}$ Triangle area = $\frac{1}{2} \times MN \times SR = \frac{1}{2} \times (36 + 4) \times \frac{400}{3}$ $= \frac{8000}{3} \text{ (units)}$	M1 M2 M3 M4 Ans
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C1)

Vertical distance (diff in y coordinate between points) $7\sqrt{10} \sin 71.57 = 21.12$ y-coord of point B = $-10 + 21 = \mathbf{11}$ Horizontal distance (diff in x-coord) $7\sqrt{10} \cos 71.57 = 6.63$ x-coord of point B = $-4 + 7 = \mathbf{3}$ $(3, 11)$	M1 Ans M2 Ans
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C2)

Gradient of line linking A & B $\frac{11 + 10}{3 + 4} = 3$ Perpendicular gradient = $-\frac{1}{3}$ Pass through B, so $y - 11 = -\frac{1}{3}(x - 3)$ $33 - 3y = x - 3$ $\mathbf{x + 3y - 36 = 0}$	M1 M2 Ans
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C3)

Using equation of circle, where (h, k) = Point O	
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$(x - h)^2 + (y - k)^2 = 185$ With point A: $(-4 - h)^2 + (-10 - k)^2 = 185$ With point B: $(3 - h)^2 + (11 - k)^2 = 185$ -> Solving by simultaneous equation (any method) $x = -8 \text{ \& } y = 3, \quad \text{or } x = 7 \text{ \& } y = -2$ Since the point lies on the right of the y-axis $= (7, -2)$	M1 M2 M3 M4 Ans
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C4)

$OA = OB = \sqrt{185}$ $AB = \sqrt{(3 - (-4))^2 + (11 - (-10))^2} = \sqrt{490}$ Find angle AOB, using Cosine Rule $490 = 2(185) - 2(185)(\cos \theta)$ $\cos \theta = \frac{12}{37}$ To change into $\sin \theta$, use any Trigonometric identity $\sin \theta = \frac{35}{37}$ Area of sector = $\frac{1}{2}ab(C)$ $= \frac{1}{2}(185) \sin^{-1}\left(\frac{35}{37}\right)$ $= \mathbf{114.746 \text{ (units)}}$	M1 M2 M3 M4 M5 Ans
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Relations - Solutions

(/35)

A1)

$a' = ae^t$ $\frac{a'}{a} = e^t, \quad \int \frac{da}{a} = \int e^t dt$ $\ln a = e^t + c$ $a = e^{(e^t+c)}$ As $a(0) = e$ $e = e^{(e^0+c)} = e^{1+c}, \quad c = 0$ $a = e^{e^t}$	M1 M2 M3 Ans
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A2)

$a' = ae^t = e^{e^t} e^t = e^{(e^t+t)}$ $a'' = \frac{da'}{dt}$ Using correct u-sub: $u = e^t + t \rightarrow \frac{du}{dt} = e^t + 1$ $\frac{da'}{dt} = \frac{da'}{du} \times \frac{du}{dt}$ $= e^u(e^t + 1) = e^{(e^t+t)}(e^t + 1)$ $= e^{(e^t+2t)} + e^{(e^t+t)}$	M1 M2 M3 Ans
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A3)

Given: $f(t) = e^t + 2t, g(t) = e^t + t$ $\frac{2f(t)}{g(t)} = \frac{2(e^t + 2t)}{e^t + t} = \frac{2e^t + 4t}{e^t + t}$ As $b(t) = 2$ $2(e^t + t) = 2e^t + 4t, \quad 2t = 0$ $t = 0$	M1 M2 M3 Ans
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B)

$y = x \ln(x^2 - 2)$ $\frac{dy}{dx} = \frac{2x^2}{x^2 - 2} + \ln(x^2 - 2)$ Using $\delta y = \frac{dy}{dx} \times \delta x$: $\delta y = \left(\frac{2x^2}{x^2 - 2} + \ln(x^2 - 2) \right) \times 0.04x$ Percentage change of y = $\frac{\delta y}{y} \times 100$ $\frac{\delta y}{y} \times 100 = \frac{\left(\frac{2x^2}{x^2 - 2} + \ln(x^2 - 2) \right)}{x \ln(x^2 - 2)} \times 0.04x \times 100$	M1 M2 M3 M4
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Correct sub in $x = 2$	
$= \frac{16}{\ln(2)} + 4$	M5 Ans

C)

1. Find straight line $r = a \sin^2 p + b$ Using points (0, 5) & (1, 10) to obtain simultaneous equations, correct solving: $5 = a(0) + b, \quad b = 5$ $10 = a \sin^2 1 + 5, \quad a = 5$ $r = 5 \sin^2 p + 5$	M1
Using $r = f(s)$ $5 \log(s + 3) = 5 \sin^2 p + 5, \quad \log(s + 3) = \sin^2 p + 1$	M2
Using $p = f(q)$, find p when $q = 2$ $p_{q=2} = 2^{2+1} = 8$	M3
Sub in $p = 8$ $\log(s + 3) = \sin^2 8 + 1 = 1.9789$	M4
Solve for s $s = 10^{1.9789} - 3$ $= \mathbf{92.24}$	M5 Ans

D)

1. Find dC/dx when angle x is 1.2 $C = 2r \sin\left(\frac{x}{2}\right) = 2\left(x^2 - \frac{1}{x}\right) \sin\left(\frac{x}{2}\right)$ $\frac{dC}{dx} = 2\left(2x + \frac{1}{x^2}\right) \sin\left(\frac{x}{2}\right) + \left(x^2 - \frac{1}{x}\right) \cos\left(\frac{x}{2}\right)$ $\left.\frac{dC}{dx}\right _{x=1.2} = 3.9952$	M1
	M2
	M3
2. Find δx $\delta r = \frac{dr}{dx} \delta x$ $\frac{dr}{dx} = 2x + \frac{1}{x^2}$ $\left.\frac{dr}{dx}\right _{x=1.2} = 2(1.2) + \frac{1}{(1.2)^2} = 3.0944$	M4
Using $\delta r = 0.1$, solve for δx $\delta r \times \frac{dx}{dr} = \delta x, \quad \frac{0.1}{3.0944} = 0.0323$	M5
3. Find δC $\delta C = \frac{dC}{dx} \delta x = 3.9952 \times 0.0323 = 0.1291$ $= \mathbf{12.91\%}$	M6 Ans

E1)

Relationship between S & h	
$\frac{1}{S} + a = bh^2$	M1
Given values of S & h	
$S = 0.2, h = 3, \quad S = 0.4, h = 2$	M2
By simultaneous equations	
$a = -\frac{1}{2}, \quad b = \frac{1}{2}$	M3
Rearranging in $S = f(h)$ form	
$\frac{1}{S} - \frac{1}{2} = \frac{h^2}{2}$	
$S = \frac{2}{h^2 + 1}$	M4 Ans

E2

OPTION 1:	
To find $\frac{dS}{dh}$, use quotient rule:	
$\frac{dS}{dh} = \frac{-4h}{(h^2 + 1)^2}$	M1
To find $\frac{dh}{d\theta}$, use implicit differentiation:	
$d \rightarrow \ln(h) + e^{-2h} = \sin \theta$	
$= \frac{1}{h} \frac{dh}{d\theta} + (-2e^{-2h}) \frac{dh}{d\theta} = \cos \theta$	M2
$\frac{dh}{d\theta} = \frac{\cos \theta}{\frac{1}{h} - 2e^{-2h}}$	M3
Using chain rule $\frac{dS}{d\theta} = \frac{dS}{dh} \times \frac{dh}{d\theta}$	
$\frac{dS}{d\theta} = \frac{-4h}{(h^2 + 1)^2} \times \frac{\cos \theta}{\frac{1}{h} - 2e^{-2h}}$	M4
At $h = 1, \sin \theta = \frac{1}{e^2}$ (Small Angle), thus, use approximation $\cos \theta \approx 1$	
$= \frac{-4}{(1 + 1)^2} \frac{1}{1 - 2e^{-2}}$	M5
Simplification to obtain final answer	
$= \frac{1}{2e^{-2} - 1}$	M6 Ans
OPTION 2:	
To find $\frac{dS}{dh}$, use quotient rule:	
$\frac{dS}{dh} = \frac{-4h}{(h^2 + 1)^2}$	

Substitute $h = 1000m = 1km$ into $\ln(h) + e^{-2h} = \sin \theta$ to get $\sin \theta = e^{-2}$, which is a small angle, hence use small angle approximation:

$$\sin \theta = \ln(h) + e^{-2h} \approx \theta$$

Now find $\frac{dh}{d\theta}$, which is the reciprocal of $\frac{d\theta}{dh}$:

$$\begin{aligned}\frac{d\theta}{dh} &= \frac{1}{h} - 2e^{-2h} \\ \frac{dh}{d\theta} &= \frac{1}{\frac{1}{h} - 2e^{-2h}}\end{aligned}$$

Using chain rule $\frac{dS}{d\theta} = \frac{dS}{dh} \times \frac{dh}{d\theta}$

$$\begin{aligned}\frac{dS}{d\theta} &= \frac{-4h}{(h^2 + 1)^2} \times \frac{1}{\frac{1}{h} - 2e^{-2h}} \\ &= \frac{-4}{(1 + 1)^2} \frac{1}{1 - 2e^{-2}} \\ &= \frac{1}{2e^{-2} - 1}\end{aligned}$$

Trigonometry – Solutions

(/45)

A1)

Make into sin & cos Only	
$= \frac{1}{\cos \theta} - \frac{\sin^2 \theta}{\cos \theta}$	M1
Use $\cos^2 \theta = 1 - \sin^2 \theta$	
$= \frac{1 - \sin^2 \theta}{\cos \theta} = \frac{\cos^2 \theta}{\cos \theta}$	
$= \cos \theta$	M2 Ans

A2)

Into single term	
$= \left(\frac{1}{\cos(x)} - \frac{\sin(x)}{\cos(x)} \right)^2 = \frac{(1 - \sin(x))^2}{\cos^2(x)}$	M1
Use $a^2 - b^2 = (a + b)(a - b)$	
$= \frac{(1 - \sin(x))(1 - \sin(x))}{1 - \sin^2(x)} = \frac{(1 - \sin(x))(1 - \sin(x))}{(1 + \sin(x))(1 - \sin(x))}$	M2
$= \frac{(1 - \sin(x))}{(1 + \sin(x))}$	M3 Ans

A3)

Combining fractions	
$\frac{\cos(3x) \cos(x) + \sin(3x) \sin(x)}{\sin(x) \cos(x)}$	M1
Using $\cos(A - B) = \cos A \cos B + \sin A \sin B$	
$= \frac{\cos(2x)}{\sin(x) \cos(x)}$	M2
Using $\sin(2x) = 2 \sin x \cos x$	
$= \frac{\cos(2x)}{\frac{1}{2} \sin(2x)} = \frac{2 \cos(2x)}{\sin(2x)}$	M3
$= 2 \cot(2x)$	M4 Ans

B1)

$\sin 15 = \sin(45 - 30) = \sin 45 \cos 30 - \sin 30 \cos 45$	
$= \frac{\sqrt{2}}{2} \left(\frac{\sqrt{3}}{2} \right) - \frac{1}{2} \left(\frac{\sqrt{2}}{2} \right) = \frac{\sqrt{6} - \sqrt{2}}{4}$	M1
By similar logic	

$\cos 15 = \frac{\sqrt{2} + \sqrt{6}}{4}$	M2
Rearranging equation $(\sqrt{3} + 1) \tan x + \sqrt{3} - 1 = 4\sqrt{2} \sin x$	M3
Times by cos $(\sqrt{3} + 1) \sin x + (\sqrt{3} - 1) \cos x = 4\sqrt{2} \sin x \cos x$	M3
Divide by $2\sqrt{2}$ $\frac{\sqrt{6} - \sqrt{2}}{4} \sin x + \frac{\sqrt{6} + \sqrt{2}}{4} \cos x = 2 \sin x \cos x$	M4
Sub-in sin/cos 15 $\sin 15 \cos x + \sin x \cos 15 = 2 \sin x \cos x$	M5
Using $\sin(A + B) = \sin A \cos B + \sin B \cos A$ $\sin(x + 15) = 2 \sin x \cos x$	M6
Using $\sin(2x) = 2 \sin x \cos x$ $\sin(x + 15) = \sin 2x$	M7
Solving, within range: $x = 15^\circ \text{ or } 55^\circ$	M8 Ans

C1)

$\sin(x) + 2 \rightarrow \sin(2x) + 2 \rightarrow \sin\left(2\left(x + \frac{\pi}{4}\right)\right) + 2 = \cos(2x) + 2$ 1. Dilation from the y-axis by a factor of $\frac{1}{2}$ 2. Translation in the positive x direction by $\frac{\pi}{4}$ OR	M1 Ans
$\sin(x) + 2 \rightarrow \sin\left(x + \frac{\pi}{2}\right) + 2 \rightarrow \sin\left(2x + \frac{\pi}{2}\right) + 2 = \cos(2x) + 2$ 1. Translation in the positive x direction by $\frac{\pi}{2}$ 2. Dilation from the y-axis by a factor of $1/2$.	M1 Ans

C2)

$\sin(x) + 2 = \cos(2x) + 2$ Use of $\cos(2x) = 1 - \sin^2(x)$ $\sin^2(x) + \sin(x) - 1 = 0$ $\sin(x) = \frac{1}{2}, -1$ For $\sin(x) = \frac{1}{2}$	M1 M2
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$x = \frac{\pi}{6}, \frac{5\pi}{6}, \quad f1\left(\frac{\pi}{6}\right) = \frac{5}{2}, f1\left(\frac{5\pi}{6}\right) = \frac{5}{2}$ For $\sin(x) = -1$ $x = \frac{3\pi}{2}, \quad f1\left(\frac{3\pi}{2}\right) = 1$ Coordinates (3) are: $\left(\frac{\pi}{6}, \frac{5}{2}\right) \& \left(\frac{5\pi}{6}, \frac{5}{2}\right) \& \left(\frac{3\pi}{2}, 1\right)$	M3, 4 M5 M6 Ans
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C3)

f1(x) volume: $\frac{225}{360} \times \pi \int_0^{\frac{\pi}{4}} (\sin(x) + 2)^2 dx = \frac{5\pi}{8} \int_0^{\frac{\pi}{4}} \sin^2(x) + 4 \sin(x) + 4 dx$ Using $\cos 2x = 1 - 2 \sin^2 x \rightarrow \sin^2 x = \frac{1 - \cos 2x}{2}$ $= \frac{5\pi}{8} \int_0^{\frac{\pi}{4}} \frac{1}{2} - \frac{\cos(2x)}{2} + 4 \sin(x) + 4 dx$ Correct integration Correct sub-in values $= 8.75$ f2(x) volume: $\frac{225}{360} \times \pi \int_0^{\frac{\pi}{4}} (\cos(2x) + 2)^2 dx = \frac{5\pi}{8} \int_0^{\frac{\pi}{4}} \cos^2(2x) + 4 \cos(2x) + 4 dx$ Using $\cos 4x = 2 \cos^2 2x - 1 \rightarrow \cos^2 2x = \frac{\cos 4x + 1}{2}$ $= \frac{5\pi}{8} \int_0^{\frac{\pi}{4}} \frac{\cos(4x)}{2} + \frac{1}{2} + 4 \cos(2x) + 4 dx$ Correct integration Correct sub-in values $= 10.87$ Difference $= \mathbf{2.12}$	M1 M2 M3, 4 M5 M6 M7, 8 M9 Ans
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C4)

<u>Area 1:</u> Correct limits & Correct equation $\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \sin(x) - \cos(2x) dx$ Correct integration Correct sub-in of values $= \frac{3\sqrt{3}}{2}$ <u>Area 2:</u>	M1 M2
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Correct limits & Correct equation	
$\int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} \cos(2x) - \sin(x) dx$	M3
Correct integration Correct sub-in of values	
$= \frac{3\sqrt{3}}{4}$	M4
Final ratio	
$= 2:1$	M5 Ans

D)

Using compound angle	
- $\sin(4x) = \sin(2.5x + 1.5x) = \sin 2.5x \cos 1.5x + \sin 1.5x \cos 2.5x$	
- $\sin(x) = \sin(2.5x - 1.5x) = \sin 2.5x \cos 1.5x - \sin 1.5x \cos 2.5x$	
SUM	
$\sin(x) + \sin(4x) = 2 \sin(2.5x) \cos(1.5x)$	M1
Using same logic →	
$\sin(2x) + \sin(3x) = 2 \sin(2.5x) \cos(0.5x)$	M2
SUM & factor	
$= 2 \sin(2.5x) (\cos(0.5x) + \cos(1.5x))$	M3
Using same logic →	
$= 4 \sin(2.5x) \cos(x) \cos(0.5x)$	M4
Finding Number of Solutions, by Components	
i) $\sin(2.5x) = 0, \rightarrow 5 \text{ solutions}$	
ii) $\cos(x) = 0, \rightarrow 2 \text{ solutions}$	
iii) $\cos(0.5x) = 0, \rightarrow 1 \text{ solution}$	
Total = 8 solutions	M5
BUT, Divided by $\cos x$	
$\cos x = 0 \rightarrow 2 \text{ solutions are Invalid}$	M6
Total sol = 8 - 2	
$= 6$	M7 Ans

Vectors - Solutions

(/32)

A1)

Prove Trapezium = prove that there is a pair of parallel lines $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BD} = -\frac{1}{3}b - 4a - b = -4a - \frac{4}{3}b$ $\overrightarrow{BC} = \overrightarrow{BD} + \overrightarrow{DC} = -4a - b - 2a - b = -6a - 2b$ For parallel, $\overrightarrow{AD} = k\overrightarrow{BC}$ $-4a - \frac{4}{3}b = \frac{2}{3}(-6a - 2b)$	M1 M2 Ans
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A2)

Let the intersection point be X. To find CX, find 2 Representations of CX: $\overrightarrow{CA} = \overrightarrow{CB} + \overrightarrow{BA} = 6a + \frac{7}{3}b, \quad \overrightarrow{CX} = \lambda\overrightarrow{CA} = 6\lambda a + \frac{7}{3}\lambda b \rightarrow (1)$ $\overrightarrow{CX} = \overrightarrow{CD} + \mu\overrightarrow{DB} = 2a + b + \mu(4a + b) = (2 + 4\mu)a + (1 + \mu)b \rightarrow (2)$ Equating parts: $A \rightarrow 6\lambda = 2 + 4\mu, \quad B \rightarrow \frac{7}{3}\lambda = 1 + \mu$ Correct solving: $\lambda = \frac{3}{5}, \quad \mu = \frac{2}{5}$ $\overrightarrow{CX} = \lambda\overrightarrow{CA} = \frac{3}{5}\left(6a + \frac{7}{3}b\right)$ $= \frac{18}{5}a + \frac{7}{5}b$	M1 M2 M3 M4 M5 Ans
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B1)

$\overrightarrow{AE} = \overrightarrow{AB} + \overrightarrow{BE}$ $\overrightarrow{BE} : \overrightarrow{EC} = 1 : z, \text{ so } \overrightarrow{BE} = \left(\frac{1}{1+z}\right)\overrightarrow{BC}$ Since $\overrightarrow{BC} = 2x - y$ $\overrightarrow{BE} = \left(\frac{2x - y}{1 + z}\right)$ Since $\overrightarrow{AB} = y$ $\overrightarrow{AE} = \overrightarrow{AB} + \overrightarrow{BE} = y + \left(\frac{2x - y}{1 + z}\right)$ $= \frac{(1+z)y}{1+z} + \frac{2x-y}{1+z}$ $= \frac{2x + yz}{1+z}$	M1 M2 M3 M4 Ans
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B2)

$\overrightarrow{AF} = \overrightarrow{AD} + \overrightarrow{DF}$	
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Since $\overrightarrow{DE}:\overrightarrow{DF} = 1:8, \overrightarrow{DF} = 8\overrightarrow{DE}$	
$\overrightarrow{AF} = \overrightarrow{AD} + 8\overrightarrow{DE}$	M1
$\overrightarrow{DE} = \overrightarrow{DA} + \overrightarrow{AE} = \frac{2x + yz}{1 + z} - x = \frac{x + yz - xz}{1 + z}$	M2
$\overrightarrow{AF} = x + 8\left(\frac{x + yz - xz}{1 + z}\right)$	M3
Resolving into components (x, y)	
$\overrightarrow{AF} = \frac{9 - 7z}{1 + z}x + \frac{8z}{1 + z}y$	M4
$\overrightarrow{AF}$ also = $k\overrightarrow{AB} = 0x + ky$, so	
$\frac{9 - 7z}{1 + z} = 0, \quad z = \frac{9}{7}$	M5
Sub-in z value	
$\overrightarrow{AF} = \frac{8z}{1 + z}y = \frac{8\left(\frac{9}{7}\right)}{1 + \frac{9}{7}}y$	M6
$= \frac{9}{2}y$	M7 Ans

C)

1. Find height of Point Y:	
- Angle that $\overrightarrow{AX}$ makes with the ground	
$= \tan^{-1} \frac{2}{9} = 12.5288^\circ$	M1
- Angle that $\overrightarrow{AY}$ makes with the ground	
$= 70 - 12.5288 = 57.4712^\circ$	M2
- i component of $\overrightarrow{AY} = 9$, j component	
$= 9 \tan 57.4712^\circ = 14.1115$	M3
- Height of Point Y	
$= 2 + 14.1115 = 16.1115$	M4
2. Solve BX & BY:	
- Let $\overrightarrow{XB} = -pi - qj$, & $\overrightarrow{BY} = -pi + rj$	
- Set equations:	
i) $3\sqrt{p^2 + r^2} = 2\sqrt{p^2 + q^2}$	M5
ii) $q + r = 16.1115$	
iii) $\frac{q}{p} \times -\frac{r}{p} = -1$	
Correct solving	
$p = 7.4361, q = 11.1541, k = 4.9574$	M6
$\overrightarrow{AB} = \overrightarrow{AX} + \overrightarrow{XB} = 9i - 2j - (-7.4361i - 11.1541j)$	
$= \mathbf{16.4361i + 9.1541j}$	M7 Ans

D)

Find $\overrightarrow{AC}$ by finding 2 representations	
$\overrightarrow{AC} = \lambda \overrightarrow{AD} = \lambda(\overrightarrow{AE} + \overrightarrow{ED}) = \lambda(3a + 3b)$	
$\overrightarrow{AC} = 3\lambda a + 3\lambda b \rightarrow (1)$	M1
$\overrightarrow{AC} = \overrightarrow{AE} + \overrightarrow{EC} = \overrightarrow{AE} + \mu \overrightarrow{EB} = 6a + b - 3\mu b$	
$\overrightarrow{AC} = 6a + (1 - 3\mu)b \rightarrow (2)$	M2
Correct solving	
$\lambda = 2, \quad \mu = \frac{4}{3}$	M3
$\overrightarrow{AC} = 3(2)(a + b) = 6a + 6b$	M4
Find $\overrightarrow{CX}$ by finding 2 representations	
$\overrightarrow{CX} = p \overrightarrow{AB} = p(6a + b - 3b)$	
$\overrightarrow{CX} = 6pa - 2pb \rightarrow (3)$	M5
$\overrightarrow{CX} = \overrightarrow{CA} + \overrightarrow{AX} = \overrightarrow{CA} + q \overrightarrow{AE} = -(6a + 6b) + q(6a + b)$	
$\overrightarrow{CX} = (-6 + 6q)a + (-6 + q)b \rightarrow (4)$	M6
Correct solving	
$p = \frac{5}{3}, \quad q = \frac{8}{3}$	M7
$\overrightarrow{CX} = 6\left(\frac{5}{3}\right)a - 2\left(\frac{5}{3}\right)b$	
$\overrightarrow{CX} = \mathbf{10}\left(a - \frac{b}{3}\right)$	M8 Ans

Logarithms - Solutions

(/47)

A1)

$\log_4 x - \log_x 32 = 3 \left(\frac{3}{2} \right) = -\frac{9}{2}$	M1
Change base: $\log_4 x - \frac{\log_4 32}{\log_4 x} = -\frac{9}{2}$	M2
Substitute $\log_4 x = u$ $u - \frac{5}{u} = -\frac{9}{2}, \quad u^2 - \frac{5}{2} = -\frac{9}{2}u, \quad 2u^2 + 9u - 5 = 0$	M3
Solving quadratic $u = -5, \frac{1}{2}$ $\log_4 x = -5, \quad \text{or } \log_4 x = \frac{1}{2}$ $x = \frac{1}{1024} \text{ or } x = 2$	M4
	M5 Ans

A2)

Change base: $= \log_x \frac{1}{2} - \frac{\log_x \frac{2}{x}}{\log_x \frac{1}{\sqrt{x}}}$	M1
Simplify denominator: $= \log_x \frac{1}{2} - \frac{\log_x \frac{2}{x}}{-\frac{1}{2}}, \quad = \log_x \frac{1}{2} + 2 \log_x \frac{2}{x} = -1$	M2
Separate terms: $\log_x \frac{1}{2} + 2 \log_x 2 + 2 \log_x \frac{1}{x} = -1$ $\log_x \frac{1}{2} + \log_x 4 - 2 = -1$ $\log_x 2 = 1$ $x = 2$	M3
	M4 Ans

B1)

Equation 1: $\log_2 \sqrt{a} + \log_2 b = \frac{5}{2}, \quad \log_2 \sqrt{a}b = \frac{5}{2}, \quad 2^{\frac{5}{2}} = \sqrt{a}b,$ $\frac{32}{a} = b^2$	M1
Equation 2:	M2

$2^a = \frac{2^{16}}{2^{2b^2}}, \quad \log_2 2^a = \log_2 2^{16-2b^2}$ $a = 16 - 2b^2$ 1 into 2: $a = 16 - 2\left(\frac{32}{a}\right)$ $a^2 - 16a + 64 = 0, \quad (a - 8)^2 = 0, \quad a = 8$ Using Eq 1 or 2, obtain b value → *MUST be Positive, as $b \geq 0$ in Log func. domain $\mathbf{a = 8, \quad b = 2}$	M3 M4 M5 M6 Ans
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B2)

Option 1: $xy^2 = 1, \quad x^2y = 2^3 = 8$ Solving: $\frac{x^2y^4}{x^2y} = \frac{1}{8}, \quad y^3 = \frac{1}{8}$ $\mathbf{y = \frac{1}{2}, \quad x = 4}$ Option 2: $\log_2 x + 2 \log_2 y = 0, \quad 2 \log_2 x + \log_2 y = 3$ By simultaneous equations: $3 \log_2 x = 6 \text{ or } 3 \log_2 y = -3$ Solving: $\mathbf{x = 4, \quad y = \frac{1}{2}}$	M1 M2 M3 Ans M1 M2 M3 Ans
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C1)

Stopping distance = thinking & braking $= 9 (\ln(t^4) + \log(t))$ Change base $= 9 (\ln(t^4) + \frac{\ln(t)}{\ln(10)})$ Fraction to power $= 9 (\ln(t^4) + \ln(t^{\frac{1}{\ln(10)}}))$ Combine logs and power -> multiple $= 9 \ln t^{4 + \frac{1}{\ln(10)}} = 9 \ln t^{4.43429}$ $\mathbf{= 40 \ln(t)}$	M1 M2 M3 M4 Ans
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C2)

$\frac{ds}{dt} = v = \frac{40}{t}$	M1
$\frac{40}{t} < 2t, \quad 20 < t^2, \quad t = \pm 2\sqrt{5}$	M2
Reject negative time, so $t = 2\sqrt{5}$	M3 Ans

C3)

$\frac{dv}{dt} = a = -\frac{40}{t^2}$	M1
$\left -\frac{40}{t^2} \right > \frac{40}{t}, \quad 40t > 40t^2, \quad t < 1 \text{ and } t \neq 0$	M2
$0 < t < 1 \text{ OR } t \in (0, 1)$	M3 Ans

D)

Understanding:

- 1st investment = 5000
- Investment doubles to 10,000, earns back \$5000, so 1st profit = 5000
- He donates half, so 1st donation = 2500
- Re-invests other 2500 $\rightarrow$ 2nd investment = 5000 + 2500 = \$7500
- Again, donates half = \$3750, and re-invests other \$3750 $\rightarrow$ 3rd investment = \$11250

D1)

Let profit per doubling (k) = P_k $\rightarrow P_k = P_{k-1} + \frac{1}{2}(P_{k-1}) = \frac{3}{2}P_{k-1}$ Meaning at each doubling, profit multiplies by $\frac{3}{2}$ $\rightarrow P_k = P_0 \left(\frac{3}{2}\right)^{k-1}$ $= 5000 \left(\frac{3}{2}\right)^{k-1} \text{ or } \frac{10000}{3} \left(\frac{3}{2}\right)^k$	M1 Ans
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D2

Let donation per doubling = D_k $D_k = \frac{P_k}{2}$ $= 2500 \left(\frac{3}{2}\right)^{k-1} \text{ or } \frac{5000}{3} \left(\frac{3}{2}\right)^k$	M1 Ans
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D3)

Sub-in $a_n = ar^{n-1}, \quad \sum a_n = \frac{a(r^n - 1)}{r - 1}$ $\sum D_k = D_0 + \frac{2500 \left(\frac{3}{2}^k - 1\right)}{\frac{3}{2} - 1}$	M1
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$= 5000 + 5000 \left(\left(\frac{3}{2} \right)^k - 1 \right)$ $\sum D_k = 5000 \left(\frac{3}{2} \right)^k$	M2 Ans
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D4)

1. Find the time taken for each doubling -> For any amount of investment x to double, using compound interest - Rate = 0.5% = 0.005 - t = time in days $2x = x(1 + 0.005)^t$ $2 = (1.005)^t$ -> Solving for t: $\ln 2 = t \ln(1.005)$ $t = \frac{\ln 2}{\ln(1.005)} = 139$ 2. Find no. of doublings needed $5000 \left(\frac{3}{2} \right)^k \geq 25000$ $k \geq \frac{\ln 5}{\ln \frac{3}{2}} = 3.969$ -> So requires 4 doublings 3. Find Total time $= 4 \times 139$ $= \mathbf{556}$	M1 M2 M3 M4 M5 Ans
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D5)

New initial amount = 10000 + 5000 = \$15000 -> So 1st donation & profit = \$7500 (instead of 5000) $= 7500 \left(\frac{3}{2} \right)^k$ Using same logic as part a $k \geq \frac{\ln \frac{10}{3}}{\ln \frac{3}{2}} = 2.9694$ -> So takes 3 doublings Using growth rate of 0.3% $t = \frac{\ln 2}{\ln(1.003)} = 231$ Total Time: 3×231 $= \mathbf{693}$	M1 M2 M3 M4 Ans
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D6)

At first 100 days, rate is 0.2%	
- Growth rate = $(1.002)^{100} = 1.22, < 2$	
➔ 1 st Doubling not achieved	M1
To complete doubling, remaining time will be at rate of 0.3%	
- Growth factor = $\frac{2}{1.22} = 1.637$	
➔ Time required to finish doubling at rate of 0.3%	
$(1.003)^t = 1.637, \quad t = 165$	M2
So, first doubling occurs at	
$100 + 165 = 265 \text{ days}$	M3
Now, remaining doublings required = $4 - 1 = 3$ more Doublings	
-> Doublings (at rate of 0.3%) will take 231 days/doubling	M4
-> So final time	
$= 265 + 3 \times 231$	M5
$= \mathbf{958}$	M6 Ans

A1)

By integration by parts	
$= [e^{-st}(y)]_0^{\infty} - \int_0^{\infty} -se^{st}(y)dt$	M1
1 st part: Sub-in bounds and simplify	
$= \left[\frac{1}{e^{\infty}}(y) - e^0(y) \right] = -y(0)$	M2
2 nd part: Recognise L(y) form	
$= s \left[\int_0^{\infty} e^{st}(y)dt \right] = s\mathcal{L}(y)$	M3
$= s\mathcal{L}(y) - y(0)$	M4 Ans

A2)

Since $L(y') = sL(y) - y(0)$	
$L(y'') = L([y'])' = sL(y') - y'(0)$	M1
$= s[sL(y) - y(0)] - y'(0)$	
$= s^2L(y) - sy(0) - y'(0)$	M2 Ans
For 3 rd order:	
$L(y''') = L([y''])' = s^2L(y') - sy'(0) - y''(0)$	
$= s^3L(y) - s^2y(0) - sy'(0) - y''(0)$	M3 Ans

B)

Sub-in Laplace transform of y differentials	M1
Sub-in initial conditions & simplify expression	
$8s^3L(y) + 77s^2L(y) - 512sL(y) - 4928L(y) - 8$	M2 Ans

C1)

$= \int_0^{\infty} e^{-st}(5e^{-9t}) dt = 5 \int_0^{\infty} e^{-(s+9)t} dt$	M1
$= 5 \times \left[-\frac{e^{-(s+9)t}}{(s+9)} \right]_0^{\infty}$	M2
$= 5 \left(-\frac{e^{-\infty}}{s+9} + \frac{e^0}{s+9} \right) = 5 \left(-0 + \frac{1}{s+9} \right)$	M3
$= \frac{5}{s+9}$	M4 Ans

C2)

$\mathcal{L}(e^{at}) = \frac{1}{s-a}$	M1 Ans
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C3)

$\sinh(kt) = \frac{1}{2}(e^{kt} - e^{-kt})$	M1
---	----

Recognising e^{at} form	
$L(\sinh(kt)) = \frac{1}{2} \left(\frac{1}{s-k} - \frac{1}{s+k} \right)$	M2
Combining fractions	
$= \frac{1}{2} \left(\frac{s+k-s+k}{(s+k)(s-k)} \right)$	M3
$= \frac{k}{s^2 - k^2}$	M4 Ans

D1)

$L(y)[8s^3 + 77s^2 - 512s - 4928] - 8 = \frac{5}{(s+9)}$	M1
Factorization	
$L(y)[s^2(8s+77) - 64(8s+77)] = 8 + \frac{5}{s+9}$	M2
$L(y)[(s^2 - 64)(8s+77)] = \frac{8s+77}{s+9}$	
Extract L(y)	
$L(y) = \frac{8s+77}{8s+77} \times \frac{1}{(s^2 - 64)(s+9)}$	M3
Using $a^2 - b^2$ identity	
$L(y) = \frac{1}{(s+8)(s-8)(s+9)}$	M4 Ans

D2)

1. Partial fractions	
$= \frac{A}{s+8} + \frac{B}{s-8} + \frac{C}{s+9}$	
-> By simultaneous equations (any method)	
$A = -\frac{1}{16}, B = \frac{1}{272}, C = \frac{1}{17}$	M1
$L(y) = -\frac{1}{16(s+8)} + \frac{1}{272(s-8)} + \frac{1}{17(s+9)}$	M2
2. Recognise general form of e^{at} c ii), then by inverse Laplace:	
$y = -\frac{1}{16}e^{-8t} + \frac{1}{272}e^{8t} + \frac{1}{17}e^{-9t}$	M3 Ans

E1)

OPTION 1:	
$L(\cos(2t) + 7 \sin(2t)) = \int_0^\infty e^{-st} \cos 2t \, dt + 7 \int_0^\infty e^{-st} \sin 2t \, dt$	M1
Calculating each term independently using integration by parts, arrive at	
$\int_0^\infty e^{-st} \cos 2t \, dt = \left[\frac{s^2}{s^2 + 4} \left(\frac{e^{-st}}{s} \right) \left(\frac{2 \sin 2t - s \cos 2t}{s} \right) \right]_0^\infty$	M2

$= \left[\frac{e^{-st}}{s^2 + 4} (2 \sin 2t - s \cos 2t) \right]_0^\infty$ $7 \int_0^\infty e^{-st} \sin 2t \, dt = \left[\frac{7e^{-st}}{s^2 + 4} (-s \sin 2t - 2 \cos 2t) \right]_0^\infty$ Sub-in bounds $\int_0^\infty e^{-st} \cos 2t \, dt = 0 - \left(\frac{1}{s^2 + 4} \right) (-s) = \frac{s}{s^2 + 4}$ $7 \int_0^\infty e^{-st} \sin 2t \, dt = 0 - \left(\frac{7}{s^2 + 4} \right) (-2) = \frac{14}{s^2 + 4}$ Sum both terms: $= \frac{s + 14}{s^2 + 4}$ OPTION 2: Recognize that $\cos(2t) = \frac{e^{2it} + e^{-2it}}{2}$ and $\sin(2t) = \frac{e^{2it} - e^{-2it}}{2i} = -i \left(\frac{e^{2it} - e^{-2it}}{2} \right)$ $L(\cos(2t) + 7 \sin(2t)) = L\left(\frac{e^{2it} + e^{-2it}}{2} - \frac{7i(e^{2it} - e^{-2it})}{2}\right)$ Group terms with e^{2it} and e^{-2it}: $= L\left(\frac{1 - 7i}{2} e^{2it} + \frac{1 + 7i}{2} e^{-2it}\right)$ Using the general form of the Laplace Transform of e^{st}: $= \frac{1 - 7i}{2(s - 2i)} + \frac{1 + 7i}{2(s + 2i)}$ Use common denominator (cross multiplication): $= \frac{(1 - 7i)(s + 2i) + (1 + 7i)(s - 2i)}{2(s - 2i)(s + 2i)}$ $= \frac{2s + 28}{2s^2 + 8}$ $= \frac{s + 14}{s^2 + 4}$	M3 M4 M5 M6 Ans
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E2)

RHS: using general form of e^{at} $L(2e^{3x}) = \frac{2}{s - 3}$ LHS: using general form of 1st & 2nd derivatives $= s^2 L(y) - sy(0) - y'(0) - 3sL(y) + 3y(0) + 2L(y)$	M1 M2
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Sub-in initial conditions	
$= (s - 1)(s - 2)L(y) - 5s + 8$	M3
Equate LHS = RHS & solve for $L(y)$	
$(s - 1)(s - 2)L(y) - 5s + 8 = \frac{2}{s - 3}$	
$L(y) = \frac{2}{(s - 1)(s - 2)(s - 3)} + \frac{5s - 8}{(s - 1)(s - 2)}$	M4
By partial fractions	
$L(y) = \frac{2}{s - 3} - \frac{2}{s - 2} + \frac{1}{s - 1} + \frac{2}{s - 2} + \frac{3}{s - 1} = \frac{2}{s - 3} + \frac{4}{s - 1}$	M5
By inverse Laplace transform	
$y = 2e^{3x} + 4e^x$	M6 Ans

Problems in 3D - Solutions

(/32)

A)

$\vec{F}_1 = ai + bj + ck, \quad \vec{F}_2 = di + ej + fk$	
Diagram 1 (X-Y axis, so NO show k parts)	M1
- $a = -d$	M2
- $b = -e$	M3
$\rightarrow \sqrt{a^2 + b^2} = \sqrt{d^2 + e^2}$	
$\rightarrow \tan 60^\circ = \sqrt{3} = \frac{b}{a}, \rightarrow b = \sqrt{3}a$	
Diagram 2 (shows the Z-axis, while X-Y become horizon)	M4
- $\alpha + 90^\circ + \beta = 180^\circ, \alpha + \beta = 90^\circ$	
- In triangles $\vec{F}_1, 180 - \beta - 90 = \alpha$	
- In triangles $\vec{F}_2, 180 - \alpha - 90 = \beta$	
$\rightarrow$ By AAA, triangles are SIMILAR	M5
Using Ratio of Lengths of Similar Triangles	
$\frac{2 \vec{F}_2 }{c} = \frac{ \vec{F}_2 }{\sqrt{a^2 + b^2}}, \quad c = 2\sqrt{a^2 + b^2}$	M6
$\frac{2 \vec{F}_2 }{\sqrt{d^2 + e^2}} = \frac{2 \vec{F}_2 }{\sqrt{a^2 + b^2}} = \frac{ \vec{F}_2 }{f}, \quad f = \frac{\sqrt{a^2 + b^2}}{2}, \quad c = 4f$	M7
Since $c + f = 10$	
$c = 8, \quad f = 2, \quad \sqrt{a^2 + b^2} = 4$	M8
Using $b = \sqrt{3}a$	
$a = 2, \quad b = 2\sqrt{3}$	M9
$\vec{F}_1 = 2i + 2\sqrt{3}j + 8k, \quad \vec{F}_2 = -2i - 2\sqrt{3}j + 2k$	
$\vec{F}_1 - \vec{F}_2 = 4i + 4\sqrt{3}j + 6k$	M10 Ans

B1)

Projected in a direction going towards the negative y direction, making a 30° angle with the x-axis, so:	
$\tan 30 = -\frac{y}{x}$	
$x = -\sqrt{3}y \rightarrow (1)$	M1
Projected in an upward direction that makes a 60° angle with the z-axis, so: (angle of elevation/angle from $z = 0$ plane = $90 - 60 = 30^\circ$)	
$\tan 30 = \frac{z}{\sqrt{x^2 + y^2}}$	M2
$3z^2 = x^2 + y^2 \rightarrow (2)$	M3

Initial velocity = 2 ms^{-1} , meaning Magnitude of vector = 2 $x^2 + y^2 + z^2 = 2^2 = 4 \rightarrow (3)$	M4
Sub in (2) into (3) $4z^2 = 4, z = \pm 1$	
Since UPWARDS direction $z = 1$	M5
Sub back into (2) $x^2 + y^2 = 3$	
Sub (1) into (2) $3y^2 + y^2 = 3, y = \pm \frac{\sqrt{3}}{2}$	
Since in NEGATIVE Y direction $y = -\frac{\sqrt{3}}{2}$	M6
Hence, $x = \frac{3}{2}$	
Velocity vector of particle is given by	M7
$\left(\frac{3}{2}i - \frac{\sqrt{3}}{2}j + k\right)/t$	M8 Ans

B2)

<u>X component:</u> Constant magnetic force of 15N, so using $F = ma$ $15 = 1(a), a = 15 \text{ ms}^{-2}$ Using $s = ut + \frac{1}{2}at^2$ or $\int v_0 + at$: - initial velocity (u) = $\frac{3}{2}$ - acceleration (a) of -15 ms^{-2} from magnet (in negative x direction) $x = \frac{3}{2}t - \frac{15}{2}t^2$	M1, 2 Ans
<u>Y component:</u> No extra forces are subjected (no acceleration), so simply moves in constant velocity towards negative y direction (as initial) $y = -\frac{\sqrt{3}}{2}t$	M3 Ans
<u>Z component</u> Using $s = ut + \frac{1}{2}at^2$ or $\int v_0 + at$: - initial velocity (u) = 1 ms^{-1} - gravity, so acceleration of -9.8 ms^{-2} - z coordinate starts at 20 $z = -\frac{49}{10}t^2 + t + 20$	M4, 5 Ans

B3)

Since Point Q lies on x-axis, $x = 0$ -> Solving for time: $0 = \frac{3}{2}t - \frac{15}{2}t^2$ $t = 0(rej.), t = \frac{1}{5}$ Thus substitute in y and z dimension: $y\left(\frac{1}{5}\right) = -\frac{\sqrt{3}}{2} \frac{1}{5} = -\frac{\sqrt{3}}{10}$ $z\left(\frac{1}{5}\right) = -4.9\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right) + 20 = \frac{5001}{250}$ Point q = $\left(0, -\frac{\sqrt{3}}{10}, \frac{5001}{250}\right)$	M1 M2 Ans M3 Ans
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B4)

To find the volume, find the sum of all x times z values (sum of cross areas along y-axis) times an infinitesimal dy to obtain the full volume $V = \int_{y_Q}^{y_P} x \cdot z \, dy$ However, as x & z functions are in terms of t, correct substitution is used $y = -\frac{\sqrt{3}}{2}t, \quad \frac{dy}{dt} = -\frac{\sqrt{3}}{2}$ $V = \left \int_{t_Q}^{t_P} x \cdot z \frac{dy}{dt} dt \right = -\frac{\sqrt{3}}{2} \left \int_0^{\frac{1}{5}} \left(\frac{3}{2}t - \frac{15}{2}t^2 \right) \left(-\frac{49}{10}t^2 + t + 20 \right) dt \right $ $= -\frac{\sqrt{3}}{2} \left \int_0^{\frac{1}{5}} \left(\frac{147}{4}t^4 - \frac{147}{20}t^3 - \frac{15}{2}t^3 - 150t^2 + \frac{3}{2}t^2 + 30t \right) dt \right $ $= -\frac{\sqrt{3}}{2} \left \left[\frac{147}{20}t^5 - \frac{147}{80}t^4 - \frac{15}{8}t^4 - 50t^3 + \frac{1}{2}t^3 + 15t^2 \right]_0^{\frac{1}{5}} \right $ $= -0.17 $ Volume is equal to 0.17 m³	M1 M2 M3 M4 M5 M6 Ans
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A1)

Write out LHS $\frac{(a+1)!}{b!(a-b+1)!}$	
Write out RHS $\frac{a!}{(b-1)!(a-b+1)!} + \frac{a!}{b!(a-b)!}$	M1
$RHS = \frac{a!}{(b-1)!(a-b+1)!} + \frac{a!}{b!(a-b)!}$ $= \frac{a!b}{b!(a-b+1)!} + \frac{a!(a-b+1)}{b!(a-b+1)!}$ $= \frac{a!}{b!(a-b+1)!} (a-b+1+b)$ $= \frac{(a+1)!}{b!(a-b+1)!} = LHS$	M2 M3 Ans

A2)

Recognize the binomial theorem:	
$(a+b)^n = \sum_{j=0}^n \binom{n}{j} a^{n-j} b^j$	M1
Substitute $a = b = 1$:	
$(1+1)^n = \sum_{j=0}^n \binom{n}{j} 1^{n-j} 1^j$ $2^n = \sum_{j=0}^n \binom{n}{j}$	M2 Ans

A3)

Consider the base case $n = 1$:	
$LHS = 1 \binom{1}{1} = 1 = 1 \cdot 2 \cdot 1^0 = RHS$	M1
Holds for $n = 1$. Assume expression is true for all $n = m$. Consider $n = m + 1$, evaluate LHS.	
Using Pascal's Identity: $\binom{m+1}{k} = \binom{m}{k} + \binom{m}{k-1}$.	M2
Thus	
$\sum_{k=1}^{m+1} k \binom{m+1}{k} = \sum_{k=1}^{m+1} k \binom{m}{k} + \sum_{k=1}^{m+1} k \binom{m}{k-1}$	M3
Let $j = k - 1$, so that	
$\sum_{k=1}^{m+1} k \binom{m}{k-1} = \sum_{j=0}^m (j+1) \binom{m}{j} = \sum_{j=0}^m j \binom{m}{j} + \sum_{j=0}^m \binom{m}{j}$	M4
According to the assumption, $\sum_{j=0}^m j \binom{m}{j} = m \cdot 2^{m-1}$	

Using result from ii, $\sum_{j=0}^m \binom{m}{j} = 2^m$ And $\sum_{k=1}^{m+1} k \binom{m}{k} = \sum_{k=1}^m k \binom{m}{k} + (m+1) \binom{m}{m+1} = m \cdot 2^{m-1} + 0$ Therefore $\sum_{k=1}^{m+1} k \binom{m+1}{k} = 2m \cdot 2^{m-1} + 2^m = 2^m(m+1)$ RHS for $n = m+1$, $RHS = 2^{m-1+1}(m+1) = LHS$, QED.	M5 M6 Ans
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B1)

Substitute in $f(x) = \ln(1+x)$ and $a = 0$ $f'(x) = \frac{1}{1+x}$ $f^{(2)}(x) = \frac{-1}{(1+x)^2}$ $f^{(3)}(x) = \frac{2}{(1+x)^3}$ $f^{(4)}(x) = \frac{-6}{(1+x)^4}$ Noticing the pattern: $f^{(n)}(x) = (-1)^{n-1}(n-1)!(1+x)^{-n}$ Substituting into Taylor: $\ln(1+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ $f^{(n)}(0) = (-1)^{n-1}(n-1)!$ Then $\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^{n-1}(n-1)!}{n!} x^n = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$ Hence $(a, b, c) = (-1, 1, 1)$	M1 M3 M4 Ans
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B2)

Start with LHS, substitute expression from bi) to here: $\int_0^r \frac{\ln(1+x)}{x} dx = \int_0^r \frac{\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n}{x} dx$ $= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \int_0^r x^{n-1} dx$ $= \sum_{n=1}^{\infty} \frac{(-1)^{n-1} r^n}{n^2}$ $= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} r^n}{n^2} = RHS$	M1 M2 M3 Ans
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B3)

From bii), substitute $r = 1$.	M1
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$I = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$	
Define a sum	M2
$S = \sum_{n=1}^{\infty} \frac{1}{n^2}$	
Separating terms with even and odd:	M3
$S_{even} = \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \frac{S}{4}$	M4
Hence $S_{odd} = S - \frac{S}{4} = \frac{3S}{4}$	
Separating terms with even and odd:	
$I_{even} = \sum_{n=1}^{\infty} \frac{-1}{(2n)^2} = \frac{-S}{4}$	
Notice that $S_{odd} = I_{odd} = \frac{3S}{4}$	
Hence	
$I = I_{odd} + I_{even} = \frac{S}{2} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$	M5 Ans

B4)

LHS is the average of x^2 in interval $[-\pi, \pi]$:	
$\frac{1}{\pi - (-\pi)} \int_{-\pi}^{\pi} x^2 dx$	
Applying Power Rule and substituting π and $-\pi$:	M1
$= \frac{1}{2\pi} \left(\frac{2\pi^3}{3} \right)$	
$= \frac{\pi^2}{3}$	M2
Evaluate:	
$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos(nx) dx$	
However, notice that $\cos(nx)$ is an even function, which means that it is symmetrical over y-axis, and multiplying by x means that the graph at $x > 0$ is mirrored (wrt x-axis) compared to the graph at $x < 0$. Hence, the areas cancel. (Mark given for any demonstration/words to that effect)	M3
$a_n = 0$	
$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$	
Using $u = x, du = dx, dv = \sin(nx) dx, v = \frac{-\cos(nx)}{n}$	M4

$\int x \sin(nx) dx = \frac{-x \cos(nx)}{n} - \int \frac{-\cos(nx)}{n} dx$ $= \frac{-x \cos(nx)}{n} + \frac{\sin(nx)}{n^2}$	M5
Now evaluating from $-\pi$ to π: $\frac{-\pi \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} - \frac{-(-\pi) \cos(-n\pi)}{n} - \frac{\sin(-n\pi)}{n^2}$ $= \frac{-2\pi \cos(n\pi)}{n}$	M6
Substitute into b_n: $b_n = \frac{-2 \cos(n\pi)}{n}$	
Know that n is a positive integer due to summation, hence rewrite cosine: $b_n = \frac{-2(-1)^n}{n}$ $= \frac{2(-1)^{n+1}}{n}$	M7
RHS of Parseval Identity: $\frac{1}{2} \sum_{n=1}^{\infty} \frac{4(-1)^{2n+2}}{n^2}$ $= \frac{1}{2} \sum_{n=1}^{\infty} \frac{4}{n^2}$	M8
Equating LHS and RHS: $\frac{\pi^2}{3} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{4}{n^2}$ $\frac{\pi^2}{12} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$	M9
Hence: $I = \frac{\pi^2}{12}$	M10 Ans

Proofs 2 – Solutions

(/38)

A1)

Pick a player M that has the maximum wins out of everyone.	M1
Assume M is not a King, this implies the existence of a player X who beat M .	M2
This would imply X has more wins than M , a contradiction to the initial statement. (Logic: think worse case, every player beaten by M also beaten by X).	M3
Hence, the person with the most wins must be the King.	M4 Ans

A2)

Assume a player M whom we know is a King, as there must be at least one King.	M1
Conditions: No one wins every game, assume M is beaten by a player X .	M2
However, only with this condition, X is king instead of M , thus assume the existence of player Y , whom M beats, yet X loses.	M3
This forms a closed loop where $M \Rightarrow Y \Rightarrow X \Rightarrow M$, hence there must be at least three kings.	M4 Ans

B1)

Consider the probability that an event happens at a small interval is an independent event. Divide time t into N equal small intervals so that:	M1
$dt = \frac{t}{N}$	
The probability that an event happens at one of these intervals is thus	M2
$pdt = \frac{pt}{N}$	
Now, consider the time interval from $t = 0$ to $t = t$, we want no events. The probability that an event NOT happening at interval dt is:	M3
$\left(1 - \frac{pt}{N}\right)$	
Hence, the probability that an event NOT happening until t is:	M4
$\left(1 - \frac{pt}{N}\right)^N$	
Thus, the probability of an event happening between t and $t + dt$ is:	M5
$\lim_{N \rightarrow \infty} \left(1 - \frac{pt}{N}\right)^N pdt$	

$= e^{-pt} p dt$ (from definition of e)	M6 Ans
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B2)

To find the average value (or expectation value) of a quantity, we must multiply the probability of a value occurring times the value itself, and then integrate over all the values. $\bar{t} = p \int_0^{\infty} t \cdot e^{-pt} dt$ Using integration by parts: $u = t, du = dt, dv = e^{-pt} dt, v = -\frac{1}{p} e^{-pt}$ $p \int t \cdot e^{-pt} dt = -te^{-pt} - \int -e^{-pt} dt$ $= -te^{-pt} - \frac{1}{p} e^{-pt}$ Integrating from 0 to infinity: $\lim_{t \rightarrow \infty} te^{-pt} = 0, \text{ exponential decay faster than linear (owtte).}$ $0e^0 + \frac{1}{p} e^0 - \lim_{t \rightarrow \infty} te^{-pt} - \frac{1}{p} e^{-\infty}$ $= 0 + \frac{1}{p} - 0 - 0$ $= \frac{1}{p}$	M1 M2 M3 M4 M5 M6 Ans
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B3)

Pick a random point in time, from part ii, the next event is on average after $\frac{1}{p}$ seconds. However, this process has no preferred direction, thus the last event was on average $\frac{1}{p}$ seconds ago. Hence, the average length is $\frac{1}{p} + \frac{1}{p} = \frac{2}{p}$, different from expectation value of waiting time.	M1 M2 M3 Ans
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C1)

Square t' and x': $t'^2 = \gamma^2(t - vx)^2 = \gamma^2(t^2 - 2tvx + v^2x^2)$ $x'^2 = \gamma^2(x - vt)^2 = \gamma^2(x^2 - 2xvt + v^2t^2)$ Using the expressions above: $t'^2 - x'^2 = \gamma^2[(t^2 - 2tvx + v^2x^2) - (x^2 - 2xvt + v^2t^2)]$	M1
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$= \gamma^2(t^2 + v^2x^2 - x^2 - v^2t^2)$ $= \gamma^2[t^2(1 - v^2) - x^2(1 - v^2)]$ $= \gamma^2(1 - v^2)(t^2 - x^2)$	M2
Since $\gamma = \frac{1}{\sqrt{1-v^2}}$, $\gamma^2 = \frac{1}{1-v^2}$, hence:	M3
$t'^2 - x'^2 = t^2 - x^2$	M4 Ans

C2)

Write the Lorentz Transformations (Frame 1 to Frame 2)	
$t_2 = \gamma_u(t_1 - ux_1)$ EQ. 1	
$x_2 = \gamma_u(x_1 - ut_1)$ EQ. 2	M1
And from frame 2 to frame 3:	
$t_3 = \gamma_v(t_2 - vx_2)$ EQ. 3	
$x_3 = \gamma_v(x_2 - vt_2)$ EQ. 4	M2
We want to write “3”s in terms of “1”s to prove the problem.	
Substitute Eq.1 and Eq.2 into Eq.3:	
$t_3 = \gamma_v[\gamma_u(t_1 - ux_1) - v\gamma_u(x_1 - ut_1)]$	M3
$= \gamma_v\gamma_u[t_1 - ux_1 - vx_1 + vut_1]$	
$= \gamma_v\gamma_u[t_1(1 + uv) - x_1(u + v)]$	M4
At the end, we get:	
$t_3 = \gamma_v\gamma_u(1 + uv) \left[t_1 - \left(\frac{u + v}{1 + uv} \right) x_1 \right]$	M5 Ans
Which is in the form of a Lorentz Transformation.	

C3)

Noticing the structure from part ii, let w be the combination of velocities:	
$\gamma = \gamma_v\gamma_u(1 + uv)$	M1 Ans
$w = \frac{u + v}{1 + uv}$	M2 Ans

C4)

Define two events in spacetime:	
Event 1: (t_1, x_1) in unprimed frame, (t'_1, x'_1) in primed frame	
Event 2: (t_2, x_2) in unprimed frame, (t'_2, x'_2) in primed frame	M1
Their time intervals are in the unprimed and primed frame respectively:	
$\Delta t' = t'_2 - t'_1$ and $\Delta t = t_2 - t_1$	
$\Delta t' = \gamma(t_2 - vx_2) - \gamma(t_1 - vx_1)$	M2
$= \gamma(t_2 - t_1 - v(x_2 - x_1))$	M3
Proper time Δt has $x_1 = x_2$, therefore	
$\Delta t' = \gamma(t_2 - t_1) = \gamma\Delta t$	M4 Ans